$\lim_{x \to a} f(x) = L$

$\lim_{x \to 1} x^2 + 1 = 2$

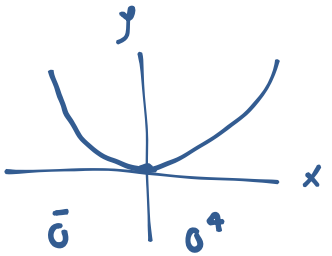
$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \frac{0}{0}$

undefined

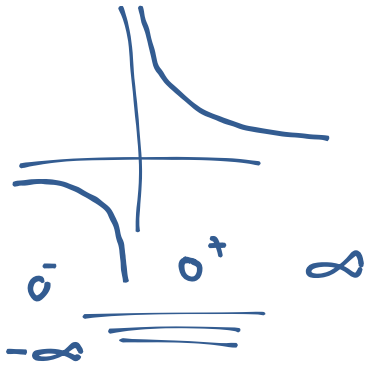
$\left[\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, \infty \cdot 0\right]$

$a, \infty, -\infty$ define

$y = x^2$



$y = \frac{1}{x}$



$\lim_{x \to a^+} f(x)$

$\lim_{x \to a^-} f(x)$

$$\lim_{x \to 2} \frac{x^2 - x - 2}{x - 2} = \frac{4 - 2 - 2}{2 - 2} = \frac{0}{0}$$

$$= \lim_{x \to 2} \frac{(x-2)(x+1)}{x-2} = 2 + 1 = 3$$

$$\lim_{x \to 1} \frac{x^2 + x - 2}{x^2 - x} = \frac{1 + 1 - 2}{1 - 1} = \frac{0}{0}$$

$$= \lim_{x \to 1} \frac{(x+2)(x-1)}{x(x-1)} = \frac{1+2}{1} = 3$$

$$\lim_{x \to 5} \frac{x^2 - 25}{x - 5} = \frac{25 - 25}{5 - 5} = \frac{0}{0}$$

$$= \lim_{x \to 5} \frac{(x-5)(x+5)}{(x-5)} = 10$$

$$\lim_{x \to 1} \frac{x^2 + x - 2}{x^2 - 1} = \frac{0}{0}$$

$$= \lim_{x \to 1} \frac{(x+2)(x-1)}{(x-1)(x+1)} = \frac{3}{2}$$

$$\lim_{x\to a}\left[f(x)\pm g(x)\right]=\lim_{x\to a}f(x)\pm\lim_{x\to a}g$$

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}$$

$$\lim_{x\to a}\sqrt[n]{f(x)}=\sqrt[n]{\lim_{x\to a}f(x)}$$

$$\lim_{x\to a}\left[f(x)\right]^n=\left[\lim_{x\to a}f(x)\right]^n$$

$f(x)=x^2$ $\qquad$ $g(x)=\frac{4}{x+1}$

$$\lim_{x\to 1}\left[f(x)+g(x)\right]$$

$$=1+2=3$$

$$\lim_{x\to 1}x^2=1$$

$$\lim_{x\to 1}\frac{4}{x+1}=2$$

$$\lim_{x\to 1}\frac{f(x)}{g(x)}=\frac{1}{2}$$

$$\lim_{x\to 1}\sqrt[4]{f(x)}=\sqrt[4]{1}=1$$

$$\lim_{x\to 1}\left[g(x)\right]^3=2^3=8$$

$x \to \infty$

$\lim_{x\to\infty} (x^2+3) = \infty + 3 = \infty$

$\infty \pm a = \infty$

$\infty * a = \infty$

$\frac{\infty}{a} = \infty$

$\frac{a}{\infty} = 0$

$\lim_{x\to\infty} \frac{x+6}{x^2+3} = \frac{\infty}{\infty}$ $\quad \div x^2 \quad \div x^2$

$= \lim_{x\to\infty} \frac{\frac{x}{x^2} + \frac{6}{x^2}}{\frac{x^2}{x^2} + \frac{3}{x^2}}$

$= \lim_{x\to\infty} \frac{\frac{1}{x} + \frac{6}{x^2}}{1 + \frac{3}{x^2}} = \frac{\frac{1}{\infty} + \frac{6}{\infty}}{1 + \frac{3}{\infty}}$

$= \frac{0+0}{1+0} = 0$

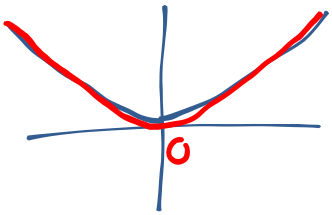
$\lim_{x\to\infty} \frac{4x^3 - 2x}{x^2 + 3x - 5} = \frac{\infty - \infty}{\infty + \infty}$ $\quad \div x^3 \quad \div x^3$

$= \lim_{x\to\infty} \frac{\frac{4x^3}{x^3} - \frac{2x}{x^3}}{\frac{x^2}{x^3} + \frac{3x}{x^3} - \frac{5}{x^3}} = \frac{4 - \frac{2}{x^2}}{\frac{1}{x} + \frac{3}{x^2} - \frac{5}{x^3}}$

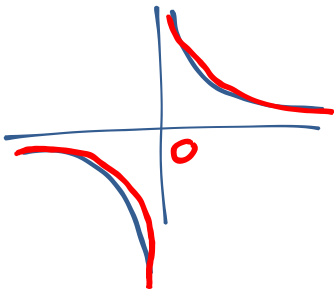
$= \frac{4 - \frac{2}{\infty}}{\frac{1}{\infty} + \frac{3}{\infty} - \frac{5}{\infty}} = \frac{4-0}{0+0-0} = \frac{4}{0} = \infty$

Continuity of Function

$y = x^2$

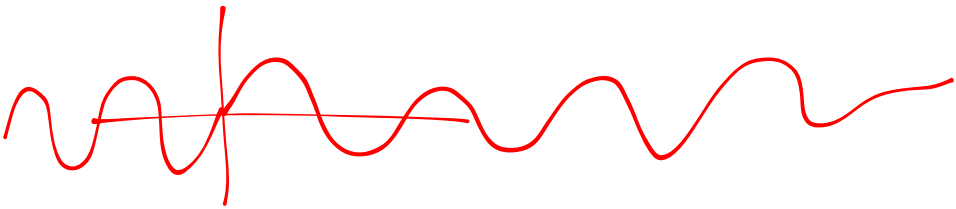


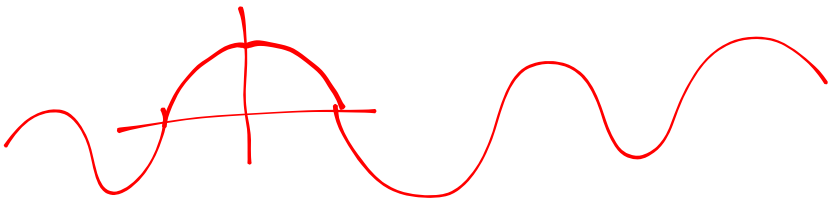
$y = \frac{1}{x}$



$$\lim_{x \to a^+} f(x) = \lim_{x \to a^-} = f(a)$$

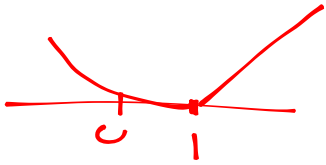
Continuous Function





$$f(x) = \begin{cases} x+2 & x > 1 \\ x^2 & x \leq 1 \end{cases}$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} (x+2) = 3$$



$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} x^2 = 1$$

$$\lim_{x \to 1^+} f(x) \neq \lim_{x \to 1^-} f(x)$$

not Continuous

$$f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x - 3} & x \neq 3 \\ 5 & x = 3 \end{cases} \quad \text{at } x = 3$$

$$\lim_{x \to 3^+} f(x) \quad \frac{x^2 - x - 6}{x - 3} = \frac{9 - 3 - 6}{3 - 3} = \frac{0}{0}$$

$$= \lim_{x \to 3} \frac{(x-3)(x+2)}{(x-3)} = 5$$

$$\lim_{x\to 3^-} \frac{x^2-x-6}{x-3} = 5$$

$$f(3) = 5$$

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^-} f(x) = f(3) = 5$$

Continuous

$$f(x) = \begin{cases} kx+6 & x > 3 \\ kx^2 & x \leq 3 \end{cases} \quad \text{at } x = 3$$

Continuous

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3} kx+6 = 3k+6$$

$$\lim_{x\to 3^-} f(x) = \lim_{x\to 3} kx^2 = 9k$$

$$3k+6 = 9k \Rightarrow k = 1$$